

Week 3 Guided Self-Study

- Implementing precise interrupts in pipelined processors
 - J.E. Smith, A.R. Pleszkun
 - Read: Section I to III
- CVA6S+: A Superscalar RISC-V Core with High-Throughput Memory Architecture
 - Riccardo Tedeschi
- Read/watch: selected paper sections and the talk
- Reflect: use the study questions on the next slide
- Focus: superscalar execution, precise interrupts, and the performance costs of preserving precise state

Study Questions

1. Why might a dual-issue processor achieve much less than twice the performance of a single-issue processor? Identify two bottlenecks discussed in the CVA6S+ talk.
2. How does forwarding between ALUs help CVA6S+ execute dependent instructions? Why might designers restrict which operations can be chained? Consider both IPC and clock period.
3. An older load causes a page fault after a younger instruction has computed an incremented loop counter. What state should the exception handler observe? Why is saving only the faulting instruction's PC insufficient?
4. In Example 3, why must the younger integer addition stall even though it is independent of the older floating-point addition? Explain how this restriction preserves precise interrupts and what performance opportunity it sacrifices.
5. A store has computed its address and data, but an older instruction may still raise an exception. Should the store update memory immediately? Using Section III-B, explain why preventing register updates alone would not preserve precise state.